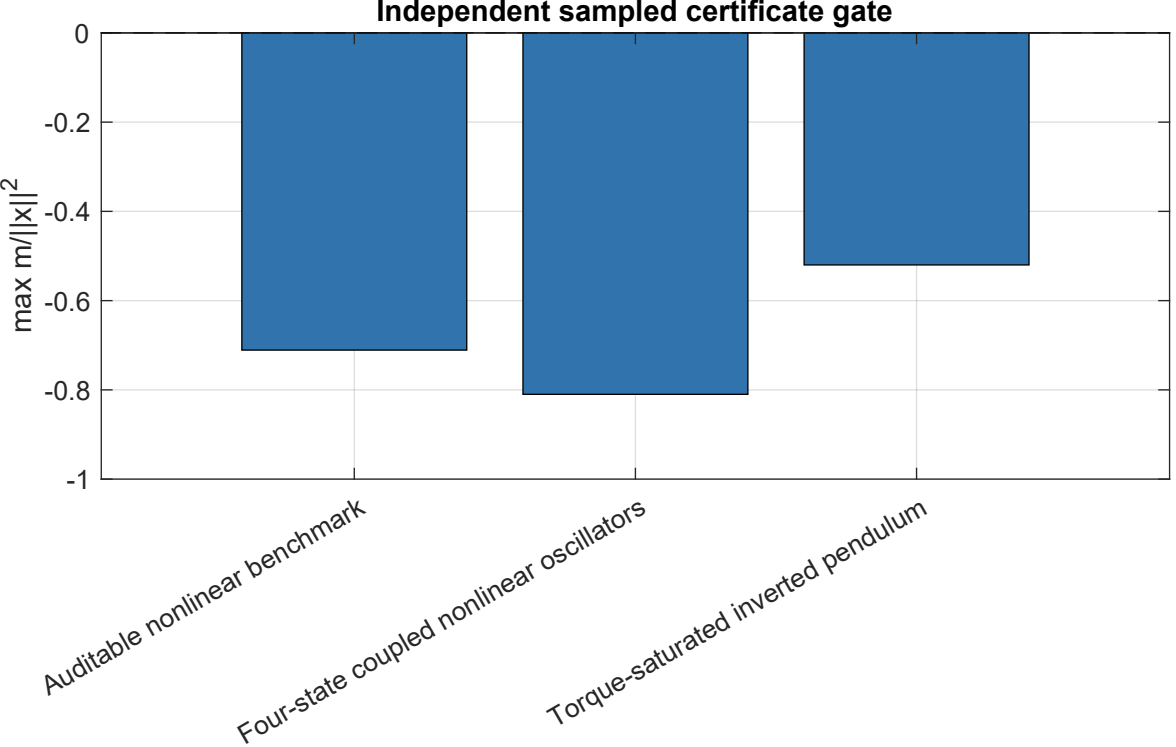
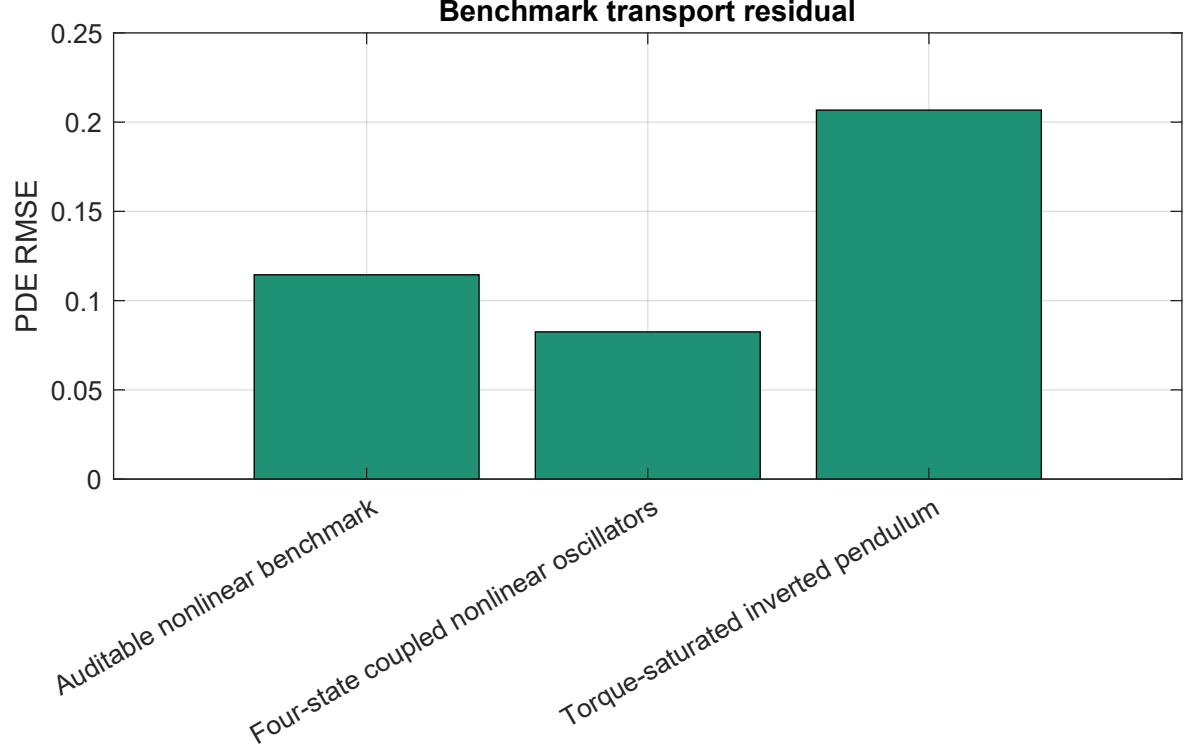
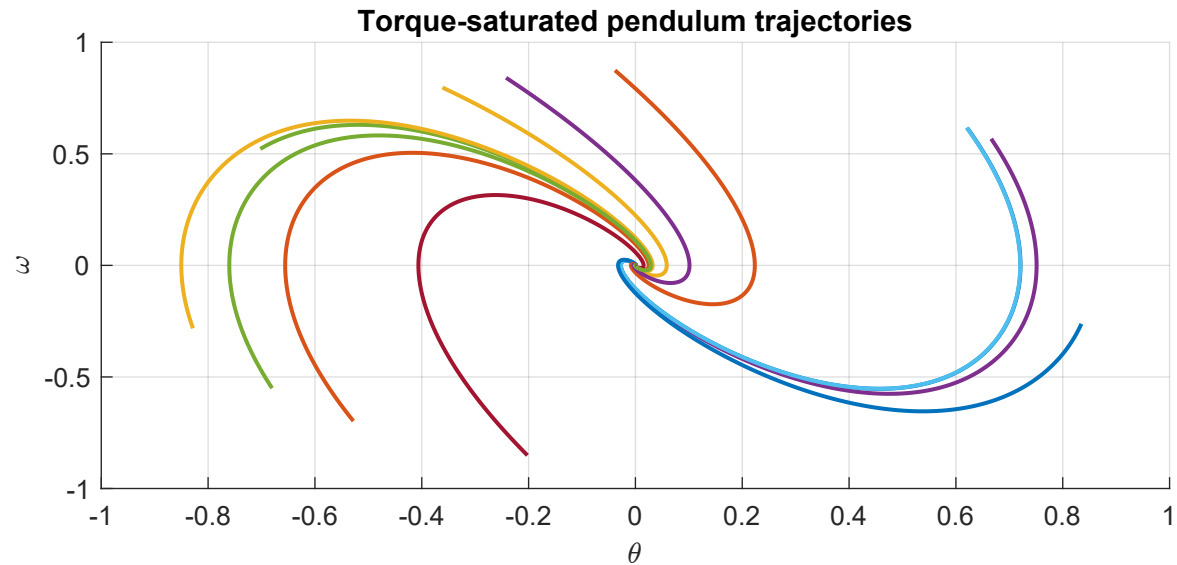
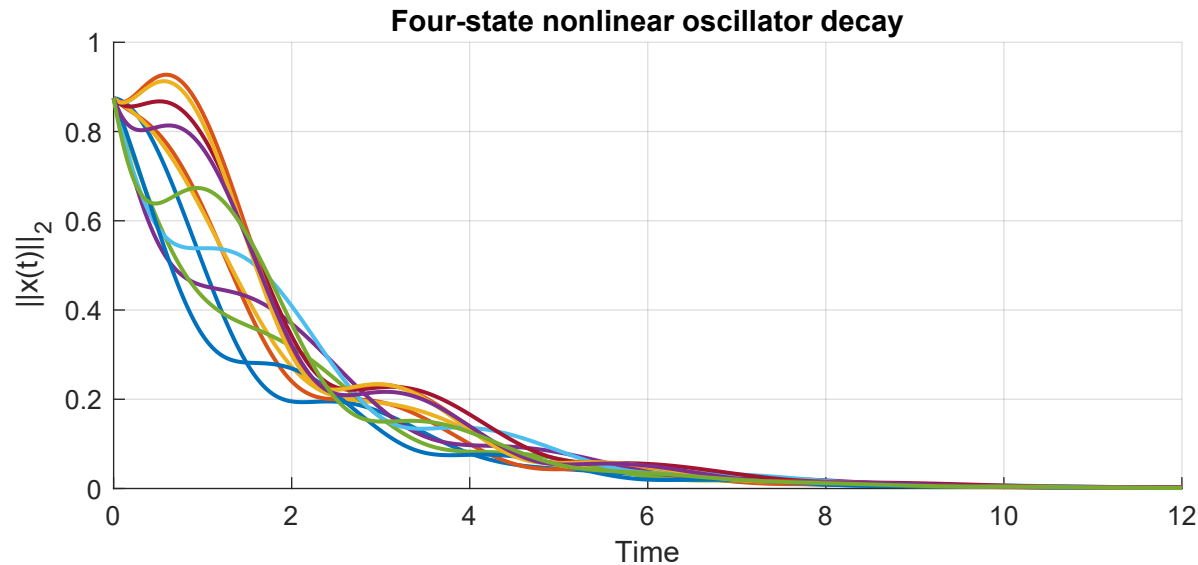

Independent sampled certificate gate
max m/||x||^2
0
-0.2
-0.4
-0.6
-0.8
-1
Auditable nonlinear benchmark
Four-state coupled nonlinear oscillators
Torque-saturated inverted pendulum


Benchmark transport residual
PDE RMSE
0.25
0.2
0.15
0.1
0.05
0
Auditable nonlinear benchmark
Four-state coupled nonlinear oscillators
Torque-saturated inverted pendulum


Torque-saturated pendulum trajectories
ω
θ
1
0.5
0
-0.5
-1
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1


Four-state nonlinear oscillator decay
||x(t)||_2
Time
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12